

Chapter 2	Configuration Space
Chapter 3	Rigid-Body Motions
Chapter 4	Forward Kinematics
Chapter 5	Velocity Kinematics and Statics
	5.1 Manipulator Jacobian
	5.2 Statics of Open Chains
Chapter 6	Inverse Kinematics
Chapter 7	Kinematics of Closed Chains
Chapter 8	Dynamics of Open Chains
Chapter 9	Trajectory Generation
Chapter 10	Motion Planning
Chapter 11	Robot Control
Chapter 12	Grasping and Manipulation
Chapter 13	Wheeled Mobile Robots

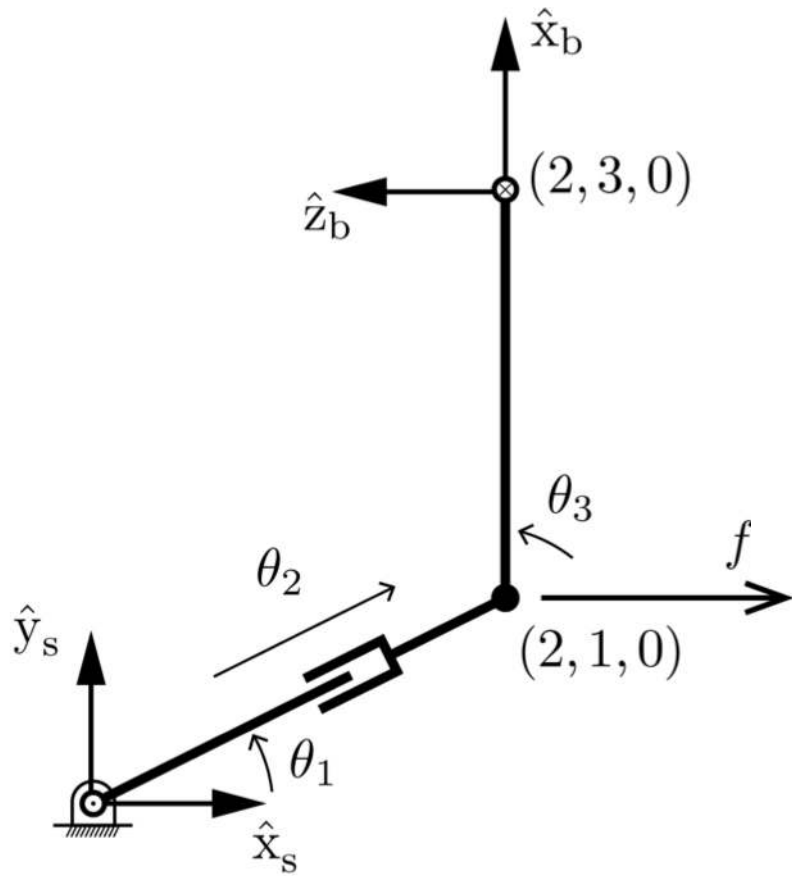
Important concepts, symbols, and equations

Robot statics: $\tau = J_*^T(\theta) \mathcal{F}_*$, where $*$ = s or b .

Proper interpretation: if a wrench $-\mathcal{F}$ is applied to the last link, then $\tau = J^T(\theta) \mathcal{F}$ is required to resist it.

If $J(\theta)$ has rank 6, then the robot can *actively* generate an end-effector wrench in any direction. The static equation is useful for force control.

If $J(\theta)$ has rank $k < 6$, then any applied wrench can be decomposed into the sum of components in k directions requiring motors to resist and components in $6 - k$ directions that are resisted by the bearings.

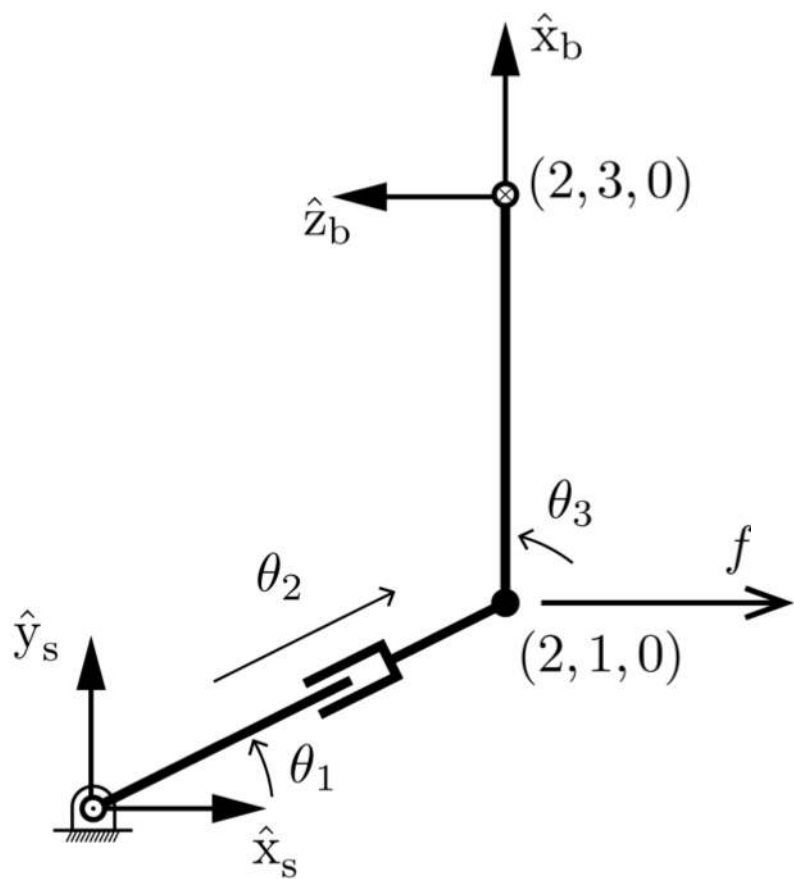


What is the 6×3 Jacobian J_b ? What is its rank?
 What wrenches can be resisted without using the motors?

$$J_b = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & -1 \\ 0 & 0 & 0 \\ 2 & 1/\sqrt{5} & 0 \\ 0 & 0 & 0 \\ 3 & -2/\sqrt{5} & 2 \end{bmatrix}$$

rank = 3
 columns are linearly independent

Any wrench out of the plane is resisted by bearings.



A linear force f to the right is applied to link 3 at the point shown. What is the corresponding wrench $-\mathcal{F}_b$? τ needed to resist it?

$$M_b = \begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ -f \end{bmatrix} = \begin{bmatrix} 0 \\ -2f \\ 0 \end{bmatrix}$$

$$-\mathcal{F}_b = \begin{bmatrix} 0 \\ -2f \\ 0 \\ 0 \\ 0 \\ -f \end{bmatrix}$$

$$\tau = J_b^T \mathcal{F}_b = \begin{bmatrix} 0 & -1 & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 1/\sqrt{5} & 0 & -2/\sqrt{5} \\ 0 & -1 & 0 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 2f \\ 0 \\ 0 \\ 0 \\ f \end{bmatrix} = \begin{bmatrix} -2f + 3f \\ -2f/\sqrt{5} \\ -2f + 2f \end{bmatrix}$$

understand
this τ ...